Library Desk API

Project Design and Technical Notes

Problem

A library service maintains a book catalogue and records loans. A member can borrow a book only when a copy is available.

Requirements

Add/list/get books; create/list loans. Reject blank titles/authors, invalid copy counts, unknown books and borrowing when stock is zero.

Architecture

app.py is the HTTP layer; LibraryService contains business rules; Database manages SQLite; Book and Loan dataclasses describe entities.

Data model

books(id PK, title, author, total_copies, available_copies). loans(id PK, book_id FK, member_name, returned). One book can have many loans.

API

GET /books; GET /books/{id}; POST /books; GET /loans; POST /loans. POST /books accepts title, author, copies. POST /loans accepts book_id and member_name.

Design notes

SQL values use parameter placeholders. Schema has NOT NULL and CHECK constraints. The demo omits authentication, returns, pagination, migrations and robust concurrency control.

Assessment discussion

Discuss HTTP/business/persistence separation, one-to-many relationships, stock consistency, returning books, authorization and simultaneous borrow requests.